

INDEPENDENT ENGINEERING CASE STUDY

# Formally verified RTL optimization of a SHA-1 core

Technical report - reproducible VTR 45 nm case study

**Four cycle-equivalent RTL rewrites improve composite PPA by 5.98%**

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Evidence snapshot: 5 August 2026

Academic open-FPGA estimate. Not ASIC signoff, a commercial FPGA result, or manufactured silicon.

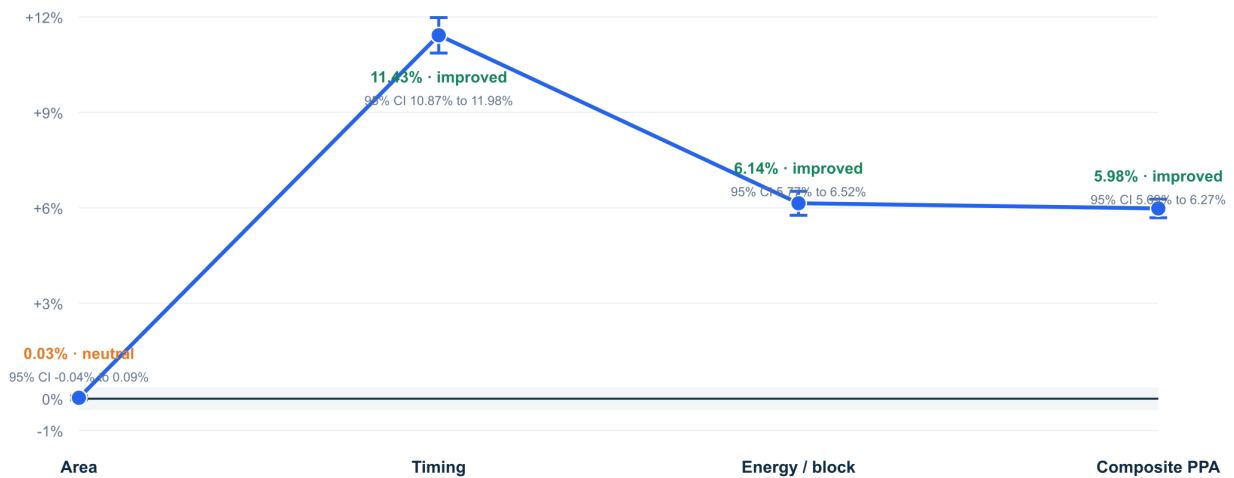
## Section 1 - Executive finding

The certified champion improves the equal-weight area-delay-energy score by 5.98% (95% CI: 5.69% to 6.27%) without changing the module interface, state-visible behavior, latency or register count.

| Metric         | Baseline median | Champion median | Paired result (95% CI)                   |
|----------------|-----------------|-----------------|------------------------------------------|
| Total area     | 16,614,693 MWTa | 16,614,693 MWTa | 0.03% better; -0.04% to +0.09% (neutral) |
| Critical path  | 15.0054 ns      | 13.28085 ns     | 11.43% better; 10.87% to 11.98%          |
| Energy / block | 12.3896 nJ      | 11.6399 nJ      | 6.14% better; 5.77% to 6.52%             |
| Composite PPA  | 1.000000        | 0.940188        | 5.98% better; 5.69% to 6.27%             |

### Certified PPA improvement vs. baseline

64 paired, fixed and search-disjoint VPR seeds · lower metric values are better



Paired log-ratio estimate. The gray band is a visual  $\pm 0.35\%$  vicinity of no change, not an acceptance threshold.

Figure 1. Certified paired improvements. Area is neutral because its interval crosses no change.

All 64 paired seeds favor the champion for timing, energy and composite score. Area produces 14 wins, 41 ties and 9 losses and is therefore reported as neutral. The result is cumulative for the four-line candidate; no exact per-line PPA attribution is claimed.

## Section 2 - Module and benchmark

The evaluated module is a sequential implementation of SHA-1 compression. It consumes a 512-bit message block through a 32-bit command/data interface and updates a 160-bit chaining state through 80 rounds of Boolean mixing, rotations, a message schedule and modulo-2<sup>32</sup> addition. A legal block occupies 80 busy cycles.

### Frozen public interface

```
module sha1(clk_i, rst_i, text_i, text_o, cmd_i, cmd_w_i, cmd_o);
  input clk_i, rst_i, cmd_w_i;
  input [31:0] text_i; output [31:0] text_o;
  input [3:0] cmd_i; output [3:0] cmd_o;
endmodule
```

Only sha.v may change. Reset behavior, command protocol, digest read order, latency, throughput and every observable cycle are fixed. This makes a candidate optimization, not a redesign of the surrounding system.

### Why SHA-1 is used

SHA-1 is a legacy benchmark, not a recommended security primitive. It is useful here because it combines sequential control, word-level arithmetic and nonlinear Boolean logic with authoritative conformance vectors. It provides a comprehensible test vehicle for the method without exposing proprietary RTL.

### Provenance and corrected reference

The starting artifact is from VTR commit 95f5c6de9e158371ba7185bf97c07a84153735d6. The upstream RTL does not produce the standard digest for abc. A separate conformance correction creates the frozen golden source. The correction is historical preparation and is not counted as an optimization.

**Corrected reference: SHA1(abc) =  
a9993e364706816aba3e25717850c26c9cd0d89d**

## Section 3 - Frozen evaluation contract

A PPA comparison is meaningful only when candidate and baseline share the same functional, physical and statistical contract. The repository freezes each item below in a machine-readable manifest and includes the exact evaluator snapshot.

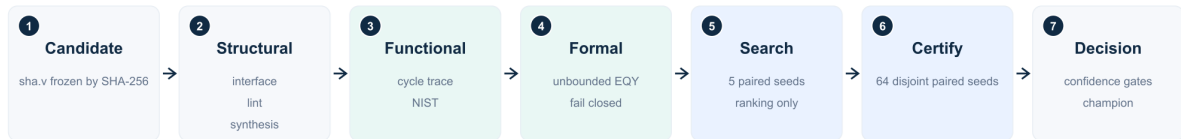
| Contract dimension  | Frozen choice                                                              |
|---------------------|----------------------------------------------------------------------------|
| Editable artifact   | sha.v only; candidate frozen by SHA-256                                    |
| Functional behavior | Exact interface, reset, protocol, latency, output order and cycle behavior |
| Conformance         | NIST SHA-1 Short and Long Message corpus; 129 cases per candidate          |
| Formal              | EQY commit 6734d8c2...; fail closed; 2 CPU, 7 GB, 10 min                   |
| Physical flow       | VTR/VPR commit 95f5c6de... under pinned Linux/amd64 image                  |
| Architecture        | k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml                                       |
| Power               | VTR PTM45, 0.9 V, 85 C; fixed 5,000-cycle active and idle traces           |
| Search              | Five exposed paired VPR seeds; non-certifying                              |
| Certification       | 64 fixed, paired and search-disjoint VPR seeds                             |
| Primary metrics     | Total MWTA, critical-path delay, energy per completed block                |

The candidate cannot alter testbenches, activity, architecture, tool flags, seeds or parsers. The evaluator, not the generator, owns acceptance. This allows the same domain to serve a hand-authored candidate, a classic evolutionary generator or an AI coding agent.

## Section 4 - Verification-first pipeline

### Correctness precedes PPA

The candidate generator is replaceable; the evaluator contract is fixed



Any failed or inconclusive correctness gate invalidates the candidate before routed PPA is spent.

Figure 2. PPA is reached only after source, functional, NIST and formal gates pass.

### Gate semantics

| Gate             | Purpose                                                              | Failure behavior                          |
|------------------|----------------------------------------------------------------------|-------------------------------------------|
| Source/interface | Reject interface drift, extra modules and simulation-only constructs | Invalid candidate                         |
| Functional       | Check reset, protocol, cycle behavior and known answers              | Invalid candidate                         |
| NIST             | 129 Short/Long conformance cases                                     | Invalid candidate                         |
| EQY              | Unbounded cycle-equivalence proof after reset                        | Fail, timeout or inconclusive invalidates |
| Search PPA       | Provisional ranking on five paired seeds                             | Never certifies                           |
| 64-seed PPA      | Fixed final evidence and acceptance                                  | Applies statistical vetoes                |

### Formal scope

The proof is conservative and structure-aware: it compares public outputs and stable internal cut points to converge within the laptop resource envelope. It supports local cycle-exact rewrites but may reject an output-equivalent candidate that substantially recodes state or retimes registers. Such rejection is a search false negative, not a false proof of correctness.

### Mutation qualification

| MCY result              | Count |
|-------------------------|-------|
| Detected by simulation  | 454   |
| Rejected only by formal | 28    |
| Proved equivalent       | 18    |
| Inconclusive            | 0     |

All 482 functionally distinguishable mutations were rejected. Mutation qualification demonstrates complementary test and formal sensitivity; it does not prove absence of all possible defects.

## Section 5 - Exact baseline-to-champion diff

```
-assign SHA1_f1_BCD = (B & C) ^ (~B & D);
+assign SHA1_f1_BCD = (B | D) & ((~B) | C);

-assign SHA1_f3_BCD = (B & C) ^ (C & D) ^ (B & D);
+assign SHA1_f3_BCD = (B & D) | (C & (B ^ D));

-assign SHA1_Wt_1 = W13 ^ W8 ^ W2 ^ W0;
+assign SHA1_Wt_1 = (W13 ^ W8) ^ (W2 ^ W0);

-assign next_A = {A[26:0],A[31:27]} + SHA1_ft_BCD + E + Kt + Wt;
+assign next_A = ({A[26:0],A[31:27]} + SHA1_ft_BCD) + (E + Kt + Wt);
```

**No register, state transition, port, command, latency or output statement changed. EQY proves the combined candidate cycle-equivalent to the corrected baseline.**

### Review principle

The transformations are deliberately reviewable. Boolean and modular arithmetic explain why each rewrite is mathematically legal; full-circuit formal equivalence is still the acceptance authority because widths, signedness, reset and sequential context can invalidate an apparently obvious source-level argument.

### Attribution boundary

The reported PPA improvement belongs to the four changes together. This campaign did not certify a full ablation matrix across 64 seeds, so it would be misleading to assign an exact percentage to any one line. The following pages explain the intent and possible synthesis consequence of each change, not a measured isolated contribution.

## Section 6 - Rewrite 1: SHA-1 choose function

Baseline:  $f1 = (B \text{ and } C) \text{ xor } ((\text{not } B) \text{ and } D)$

Champion:  $f1 = (B \text{ or } D) \text{ and } ((\text{not } B) \text{ or } C)$

The baseline terms are mutually exclusive because B and not-B cannot both be one, so their XOR equals OR. Applying distributive and consensus identities produces the champion's product-of-sums form. Both implement the SHA-1 choose behavior: C is selected when B is one, otherwise D is selected.

| B | C | D | Choose result |
|---|---|---|---------------|
| 0 | 0 | 0 | 0             |
| 0 | 0 | 1 | 1             |
| 0 | 1 | 0 | 0             |
| 0 | 1 | 1 | 1             |
| 1 | 0 | 0 | 0             |
| 1 | 0 | 1 | 0             |
| 1 | 1 | 0 | 1             |
| 1 | 1 | 1 | 1             |

The alternative form exposes different two-input operations to ABC and the LUT mapper. Better mapping is not universal; it is specific to the frozen target and the combined context.

## Section 7 - Rewrite 2: factored majority

Baseline:  $f3 = (B \text{ and } C) \text{ xor } (C \text{ and } D) \text{ xor } (B \text{ and } D)$

Champion:  $f3 = (B \text{ and } D) \text{ or } (C \text{ and } (B \text{ xor } D))$

This is the three-input majority function used in SHA-1 rounds 40 through 59. If B equals D, their value is already the majority and B xor D is zero. If B differs from D, C decides the majority. The champion writes that case split as a factored Boolean network.

| B | C | D | Majority result |
|---|---|---|-----------------|
| 0 | 0 | 0 | 0               |
| 0 | 0 | 1 | 0               |
| 0 | 1 | 0 | 0               |
| 0 | 1 | 1 | 1               |
| 1 | 0 | 0 | 0               |
| 1 | 0 | 1 | 1               |
| 1 | 1 | 0 | 1               |
| 1 | 1 | 1 | 1               |

The factorization can reduce logic depth or routing pressure after technology mapping even when a pre-mapping node count does not decrease. The representative netlist evidence later in this report supports that cumulative topology interpretation.



## Section 8 - Rewrites 3 and 4: datapath grouping

### Balanced message-schedule XOR

Baseline:  $W13 \text{ xor } W8 \text{ xor } W2 \text{ xor } W0$

Champion:  $(W13 \text{ xor } W8) \text{ xor } (W2 \text{ xor } W0)$

Bitwise XOR is associative. Parentheses make two parallel first-level operations explicit before the final XOR. A synthesizer is free to rebalance the baseline, but RTL grouping can influence intermediate optimization, LUT packing and routing topology.

### Reassociated 32-bit accumulator

Baseline:  $R + f + E + K + W$

Champion:  $(R + f) + (E + K + W)$

All operands and the destination are 32-bit vectors. The assignment therefore computes modulo  $2^{32}$ , where addition is associative. The champion exposes two partial-sum groups before their final combination. This can alter inferred adder structure and reduce an implementation dependency chain without changing any result bit.

**Widths matter: this equivalence relies on fixed 32-bit unsigned vector semantics and final modulo- $2^{32}$  truncation. Full-circuit EQY confirms the actual Verilog interpretation.**

## Section 9 - Search campaign and finalist selection

### Search progression and certification correction

Five-seed ranking guides search; only fixed 64-seed evidence selects the champion



Figure 3. Search leaders use five seeds; only the fixed 64-seed pool can certify a finalist.

| Campaign fact                 | Value               |
|-------------------------------|---------------------|
| Generations                   | 20                  |
| Submissions                   | 46                  |
| Formal pass                   | 41                  |
| Formal fail                   | 1                   |
| Rejected before formal        | 4                   |
| Unique formal-pass candidates | 29                  |
| Certified champion            | g15-wt-balanced-xor |

Generation 16 appeared best on the exposed five-seed ranking pool. The fixed certification pool revealed a statistically supported area regression, so the acceptance rule vetoed it. Generation 15 was the unique finalist to satisfy correctness, formal, metric and composite gates.

This separation prevents search overfitting from becoming a customer claim. Search can remain fast and exploratory; certification remains fixed, expensive and independent of the candidate-generation loop.

## Section 10 - Physical proxy and power model

### Open 45 nm FPGA target

The target is VTR's homogeneous k6\_N10\_I40\_Fi6\_L4\_frac0\_ff1\_45nm architecture: six-input LUTs clustered ten per logic block, with the architecture's routing and timing model. Power uses VTR PTM45 properties at 0.9 V and 85 C. MWTa is a minimum-width-transistor-area abstraction, not a physical square-micrometer area.

### Fixed activity

Two 5,000-cycle ACE traces are frozen and hashed. The active trace continuously executes diverse legal blocks and completes 60 blocks; the idle trace keeps the clock active after reset with an inactive interface. Port order is checked against the synthesized BLIF. Active and idle analyses reuse the same placement and route.

### Energy metric

**Energy per block = active total power x routed workload time / 60 completed blocks**

This metric measures modeled energy to complete the fixed corpus. It is intentionally different from instantaneous active power and from idle power. All three remain visible in the evidence so that a favorable energy score cannot conceal an unfavorable power trade-off.

### Runtime envelope

The Linux/amd64 container is capped at two CPUs, 7 GB RAM and 512 processes with runtime networking disabled. The champion's complete certification archive reports 1,091.82 seconds elapsed on the qualified Apple Silicon laptop. Host contention affects wall time, not the frozen seed identities or parsed measurement contract.

## Section 11 - Statistical method and score

Candidate and baseline use the same 64 VPR seeds. Pairing removes much of the unrelated between-seed placement/routing variation. Ratios are analyzed in log space because PPA comparisons are multiplicative and because the geometric mean preserves reciprocal symmetry.

```
Per-seed composite:  $r = \text{cube\_root}(r_{\text{area}} \times r_{\text{delay}} \times r_{\text{energy}})$ 
```

```
Published estimate:  $\exp(\text{mean}(\log(r_{\text{seed}})))$ 
```

Two-sided 95% Student-t intervals use 63 degrees of freedom. The sample size and stopping rule were declared before finalist inspection and were not extended after seeing any result. The baseline is exactly 1.0 in a paired ratio by construction; its absolute seed-to-seed variability is shown separately in the public figures and raw records.

### Acceptance

- All functional, NIST, synthesis, formal, route, power and integrity checks pass.
- At least one primary metric has a material improvement.
- No primary metric has a statistically supported regression.
- The composite upper one-sided 95% bound is below 1.0.
- Worst-case and evidence-integrity gates remain within the frozen policy.

The scalar score is a search and ranking aid, not a replacement for engineering review. Area, delay, energy, power, resources and uncertainty are always reported alongside it.

# Section 12 - Certified 64-seed result

## Every paired certification seed

Candidate improvement relative to the same baseline seed · right is better

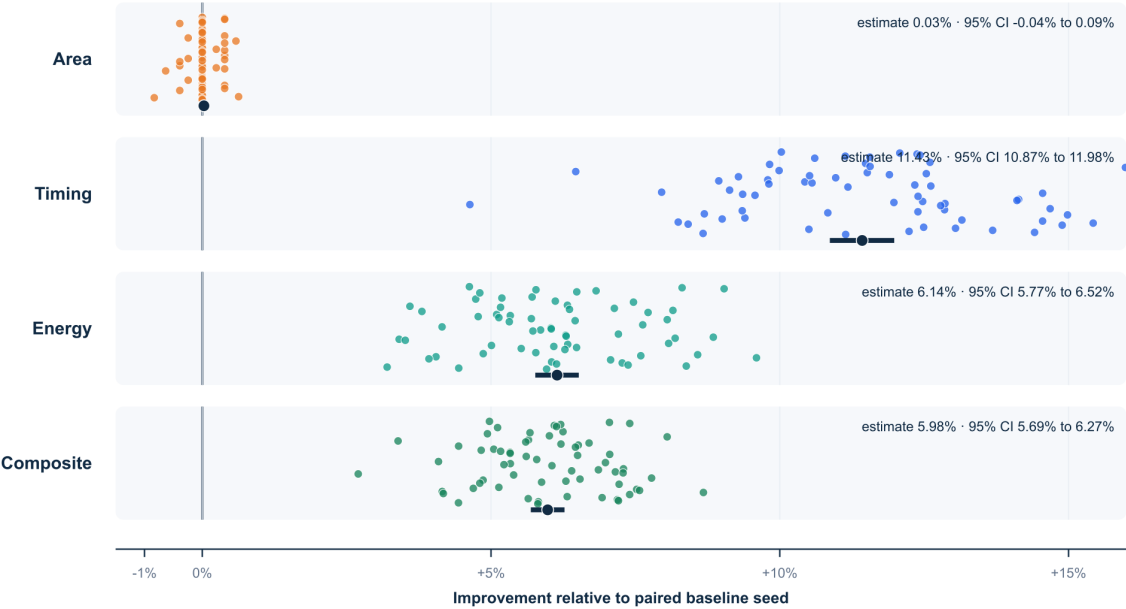


Figure 4. Paired seed distributions. Each mark compares candidate and baseline under the same VPR seed.

| Metric         | Baseline median | Champion median | Paired result (95% CI)                   |
|----------------|-----------------|-----------------|------------------------------------------|
| Total area     | 16,614,693 MWTA | 16,614,693 MWTA | 0.03% better; -0.04% to +0.09% (neutral) |
| Critical path  | 15.0054 ns      | 13.28085 ns     | 11.43% better; 10.87% to 11.98%          |
| Energy / block | 12.3896 nJ      | 11.6399 nJ      | 6.14% better; 5.77% to 6.52%             |
| Composite PPA  | 1.000000        | 0.940188        | 5.98% better; 5.69% to 6.27%             |

| Metric    | Wins / ties / losses |
|-----------|----------------------|
| Area      | 14 / 41 / 9          |
| Timing    | 64 / 0 / 0           |
| Energy    | 64 / 0 / 0           |
| Composite | 64 / 0 / 0           |

The paired estimate is the inferential headline. A simple ratio of medians gives similar but not identical values because the median of ratios is not generally the ratio of medians.

## Section 13 - Power-throughput trade-off

| Informative metric   | Baseline median | Champion median | Raw median change |
|----------------------|-----------------|-----------------|-------------------|
| Fmax                 | 66.6426 MHz     | 75.2964 MHz     | +12.99%           |
| Active total power   | 9.9125 mW       | 10.4900 mW      | +5.83%            |
| Active dynamic power | 5.0713 mW       | 5.7018 mW       | +12.43%           |
| Active static power  | 4.8394 mW       | 4.7916 mW       | -0.99%            |
| Idle total power     | 4.4670 mW       | 4.6550 mW       | +4.21%            |
| Energy / block       | 12.3896 nJ      | 11.6399 nJ      | -6.05%            |

**The champion consumes more active power per unit time but less modeled energy per completed block because it finishes the workload sooner.**

This is a genuine trade-off, not a contradiction. The declared objective values energy per operation and timing while constraining area. A product constrained by peak power, thermal density or idle power could reject the same candidate or use a different score. Customer acceptance criteria must therefore be declared before search.

The evidence retains total, dynamic and static active power plus idle power so that the composite score cannot hide this design choice.

## Section 14 - Representative netlist evidence

### What changed after synthesis and routing?

Representative seed 20 · same architecture and channel width

|                            | Baseline   | Champion          |                |
|----------------------------|------------|-------------------|----------------|
| <b>Timing graph levels</b> | 46 levels  | <b>42 levels</b>  | <b>-8.70%</b>  |
| <b>Critical path</b>       | 14.9802 ns | <b>13.4066 ns</b> | <b>-10.50%</b> |
| <b>CLB blocks</b>          | 188 CLB    | <b>187 CLB</b>    | <b>-0.53%</b>  |
| <b>ABC .names nodes</b>    | 1643 nodes | <b>1652 nodes</b> | <b>+0.55%</b>  |

The champion uses slightly more mapped logic nodes but packs into one fewer CLB and reduces the routed timing depth.

Figure 5. Seed 20 illustrates how the cumulative source change affects mapped and routed structure.

The champion has nine more ABC .names nodes, yet packs into one fewer CLB and reduces the timing graph by four levels. This rules out a simplistic claim that fewer generic mapped nodes caused the result. The better interpretation is that the new topology gives VTR a more favorable packing and routing solution.

Seed 20 is illustrative, not the statistical proof. Repeatability comes from the 64 paired implementations, in which timing, energy and composite all favor the champion.

## Section 15 - Reproducibility

### Fast evidence audit

```
make verify
```

The audit checks the RTL identities, formal and NIST evidence references, the exact 64-seed pairing, all metric log-ratio estimates, confidence intervals and the decision rule. It needs Python 3 only and performs no new EDA measurement.

### Full pinned rerun

```
./reproduce/build-image.sh
```

The build checks out VTR recursively at the pinned commit and constructs the exact Linux/amd64 evaluator image. Candidate certification runs in a new bind-mounted results directory with networking disabled. Generated failures remain failures; JSON is never edited into success.

### Evidence archive

The complete champion evidence is distributed as a path-sanitized release asset because its 70 MB compressed size and thousands of intermediate files are unsuitable for normal Git history. Its embedded PUBLIC\_SANITIZATION.json maps the 67 redacted command records from original to public member hashes. The certified original archive remains identified by SHA-256  
9983b1fef4509b9a9a592af8134be39eaa7545e5269ac7332206e86db7cce3e8.

### Clean-room expectation

A valid independent rerun starts from a clean checkout, builds or retrieves the pinned image, writes to a new evidence directory and uses the fixed certification seed pool. The public compact JSON is an auditable summary, not a substitute for the full archive when reviewing tool logs.



## Section 16 - Limitations and transfer

### Claims boundary

- Academic VTR open-FPGA estimate; not a commercial FPGA, ASIC, signoff or silicon measurement.
- No extracted parasitics, PVT corners, clock-tree signoff, IR drop, EM, DRC/LVS, package, yield or production test.
- Formal equivalence is conservative and structure-aware; it may reject legal microarchitectural redesigns.
- The 64-seed confidence interval models implementation randomness only, not process, voltage, temperature, workload or tool-version uncertainty.
- No exact PPA contribution is assigned to an individual rewrite without ablation evidence.
- SHA-1 is a legacy benchmark and must not be presented as a modern security recommendation.

### Transferable customer-pilot method

The commercial value is the evaluator architecture: freeze the customer's golden behavior and implementation context; hash every candidate; prove correctness before PPA; rank on a small paired pool; certify finalists on a fixed independent pool; and publish raw metrics, uncertainty, trade-offs and rejected finalists. A proprietary pilot would substitute customer-owned RTL, formal strategy, libraries, constraints, activity and signoff reports for VTR.

**The optimization generator may be an engineer, evolutionary system or coding agent. Acceptance remains owned by the frozen customer-specific evaluator.**

## Section 17 - Conclusion and artifact identities

A four-line, cycle-equivalent RTL change produced a repeatable 5.98% composite improvement under the declared VTR 45 nm contract. Timing and energy improved in every paired seed; area remained neutral. The result is small, reviewable, formally proved and statistically explicit.

| Artifact                 | SHA-256                                                          |
|--------------------------|------------------------------------------------------------------|
| Corrected baseline sha.v | 191a4f2148a4efda7aadd24480eb13d78a1d2c0c7e8a3fcc37c44f6a8e8011e5 |
| Champion sha.v           | 743e6c9ffcca6f00d35d5e73ba6f6478a9133a0c55a471c16d6e59d831aeeabc |
| EQY PASS marker          | ba3b47c5fbb189844a827ae395e816024c967b83444a06eccb71f9e34498ab07 |
| EQY driver log           | 203f2ab1a1db8aadcbdbf5be88e5478b1abcb51e213e941557889e1474b9cbce |
| Full evidence archive    | 9983b1fef4509b9a9a592af8134be39eaa7545e5269ac7332206e86db7cce3e8 |

### Primary references

- VTR/VPR: <https://docs.verilogtorouting.org/en/latest/vpr/>
- VTR power estimation: [https://docs.verilogtorouting.org/en/latest/vtr/power\\_estimation/](https://docs.verilogtorouting.org/en/latest/vtr/power_estimation/)
- EQY: <https://github.com/YosysHQ/eqy>
- NIST secure hashing validation:  
<https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program/secure-hashing>